

Title: Path-based and Whole Network Measures

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Path-based and Whole Network Measures

Synonyms

Centrality measures. Structural and locational properties.

Glossary

- Betweenness centrality: a measure of the proportion of shortest paths in a network passing through a specific node or link.
- Closeness centrality: a measure of how close a node is to all the other nodes of a network.
- Clustering coefficient: a measure of how often the neighbours of a node are also neighbours.
- Diameter: the maximum distance between two nodes.
- Direct connection: an edge between two nodes, usually indicating the existence of a specific relationship, e.g., a friendship between two individuals.
- Dyad: a group of two people.

- Geodesic distance (or distance): length of one of the shortest paths between two nodes.
- Indirect connection: a path between two nodes that are not directly connected through an edge.
- Node: an entity in a network, usually representing an individual.
- Path: a sequence of edges sharing common endpoints, e.g., an edge between n_i and n_j followed by an edge between n_j and n_k .
- Triangle: three nodes with an edge between every pair of them.

Definition

Path-based measures associate a value to every node in a network according to its direct and indirect connections to other nodes. For example, given a node we can compute the maximum distance to all other nodes: this measure is called node eccentricity. Whole-network measures associate a value to an entire network, providing a summary of its structure. For example, the diameter of a network is the maximum eccentricity of its nodes, and represents a global measure of the efficiency of information dissemination in that network. In this essay we cover a selection of path-based and whole-network measures.

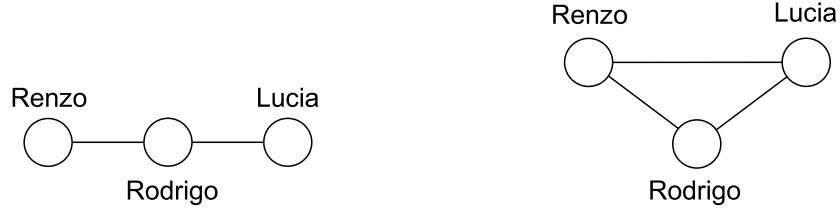
Introduction

Graphs are a widely used abstract representation of the structure of social networks, where nodes represent individuals and edges indicate relationships between them, e.g., communication acts or friendship ties. While a graph representation hides many details of the original social network—the content of communication relationships, personal data about the individuals and so on—the structure of the graph may highlight some relevant

features. What are the fastest routes that some information originated at a node can travel to reach other nodes? How important is a specific node in facilitating or slowing down information diffusion? Which one of two given networks is more efficient in disseminating information? To answer these and similar questions we need quantitative measures describing the network structure, and efficient algorithms capable of computing these measures. In this essay we describe the main graph measures related to *paths* between nodes (eccentricity, closeness, betweenness, clustering coefficient) and those used to summarize whole networks (diameter, network closeness, network clustering coefficient, and density).

To introduce the concepts discussed in this essay, we consider two simple social groups thoroughly studied in sociology: dyads (pairs of individuals) and triads (groups of three individuals). When two individuals are connected to each other we have the simplest possible social group, and flows of information between these two individuals can be fast and easy to achieve. As an example consider Figure 1(a), where nodes (circles) indicate individuals and an edge (line) between two nodes indicates mutual knowledge of e-mail addresses: if Rodrigo wants to send an e-mail to Lucia he can do that directly. Now consider Renzo: he is not connected to Lucia but he knows Rodrigo, so he can still send a message to Lucia through the common friend. We may state that there is a potential communication channel between Renzo and Lucia, but delivering a message may take longer and has a lower probability of success because Rodrigo might decide not to pass the message along. If we add another node in the chain, e.g., Agnese knows Renzo who knows Rodrigo who knows Lucia, information exchange between Agnese and Lucia is possible but may take even longer and be less likely to succeed.

From this simple example we can see how a graph induces a notion of *distance* between nodes: Rodrigo and Lucia are closer to each other than Renzo and Lucia. Distance can be considered a good indicator of how easy or fast it is to send informa-



(a) Two connected dyads (Renzo-Rodrigo and Rodrigo-Lucia)
 (b) A fully connected triad

Fig. 1. Simple social groups at the basis of many single-node and whole-network measures

tion from one node to another, and is used to define several well-known single-node measures: *eccentricity* (the maximum distance from one node to all other nodes in the network), *closeness* (the inverse average distance from one node to the others) and *betweenness* (how much information we expect to flow through a node).

While the distance between pairs of nodes and the role of nodes *in between* are useful to characterize information propagation, triads (sets of three nodes) are important to study the tendency of members of a network to form groups. In real social networks, when a node is connected to two other nodes, we observe that those two nodes are often connected to each other as well. In our example, we may expect that Renzo and Lucia also exchange their e-mail addresses after some communication mediated by Rodrigo, therefore allowing direct communication and forming a *triangle* (Figure 1(b)). A specific measure called *clustering coefficient* indicates how frequently pairs of neighbors are neighbors themselves.

Single-node metrics can often be used to compute whole-network measures by taking into consideration the contribution of all nodes. *Network closeness* and *network clustering coefficient* can be defined as averages of the corresponding single-node measures. The *diameter* of a network corresponds to the maximum eccentricity, and thus represents the maximum distance between nodes. Finally, the network *density* indicates how many of all possible connections actually exist.

Key Points

- Graphs are a widely used abstraction of the structure of social networks; for this reason, properties of a social network are inferred from those of its graph representation.
- Path-based and single-node metrics provide a quantitative measure of properties of individual nodes in a network, such as the level of “importance” of a node, whether a node is “well connected” to other nodes, and so on.
- Whole-network measures summarize attributes of the whole graph.
- In some cases, whole-network measures can be obtained by combining the values of single-node metrics over all nodes.
- Analysis of large graphs can be computationally challenging.

Historical Background

The foundations of graph theory date back to year 1735 (Alexanderson, 2006), when the Swiss mathematician Leonhard Euler solved a problem known as “the seven bridges of Königsberg”. The Prussian city of Königsberg (now Kaliningrad) was located on a bifurcation of the Pregel river where an island was present. Seven bridges were placed across the banks formed by the river and the island. The problem was stated as follows: does there exist a path that allows one to cross all bridges exactly once? Euler proved that no such path exists.

One of the first applications of graphs to the study of social interactions is generally attributed to American psychiatrist of Romanian origins Jacob Moreno (Newman, 2018). In his 1934 book *Who shall survive* (Moreno, 1934), Moreno shows and discusses diagrams of human interactions, that he calls *sociograms*, as directed or undirected graphs. Since then, Social Network Analysis (SNA) has been applied to the

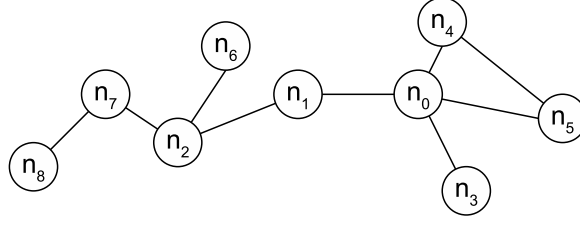


Fig. 2. Working example: a social network with nine nodes and nine edges

most diverse areas such as friendship patterns in communities, romantic and sexual relationships, collaboration of scientists, actors or musicians, networks of terrorists, and food chains in ecological systems. We will briefly discuss some of them in Section “Key applications”; the interested reader is referred to Newman (2018) for a more complete list of references.

Measures

Preliminaries: networks

In this essay, we will mostly consider the representation of a social network as an undirected, unweighted graph.

Definition 1 (Social Network). *A Social Network is an undirected graph $G = (V, E)$ where V is a set of nodes (individuals) and $E \subseteq V \times V$ is a symmetric relation indicating social connections.*

Figure 2 shows a small example with nine individuals (nodes) and nine connections edges.

In the following, we denote with n the number of nodes, and with m the number of edges (or links) of G . Given a graph G , a path $p = \langle v_0, \dots, v_k \rangle$ of length k from node u to node v is a sequence of nodes $v_0, \dots, v_k$ such that $v_0 = u$, $v_k = v$, and every consecutive pair of nodes is connected by an edge: $(v_i, v_{i+1}) \in E$, for all $i = 0, \dots, k-1$. Two nodes u and v are *connected* if there exists at least one path between them. Note

that, since edges in undirected graphs can be traversed in both directions, a path from u to v can always be reversed to obtain a path from v to u .

Representing a social network as an undirected, unweighted graph is quite common; however, it is important to keep in mind that such representation is an abstraction based on a set of assumptions, which may or may not hold.

The first assumption is to consider *undirected* graphs, where edges can be traversed in both directions. Undirected graphs are an appropriate representation of social networks where relationships between individuals are symmetric. For example, “friendship” in Facebook is symmetric. However, the following/follower relation on the X (formerly, Twitter) network is not symmetric, therefore a better representation would be based on *directed* graphs: if Lucia is following Renzo on X/Twitter, Renzo is not necessarily following Lucia. Many concepts defined for undirected graphs still hold for the directed case, such as the definition of geodesic path used as a basis for most of the measures presented in this essay. However, in directed graphs there can be a path from node u to node v without any path from v to u . In this case the distance between v and u is infinite, which may require adjustments in those measures which are based on maximizing or averaging distances. Extensions for directed graphs are presented by White and Borgatti (1994) and can be found in SNA textbooks (Wasserman and Faust, 1994; Borgatti et al, 2018).

The second assumption is to consider *unweighted* graphs, where edges do not carry any “cost” or “weight” associated with them. Adding weights to edges could be useful to convey additional details on the underlying social network. For example, the weight of an edge may represent the strength of the social relationship it represents. This kind of weight can be computed by looking at the communication acts, e.g., by counting the number of messages exchanged between two nodes. Extensions of path-based measures to weighted graphs are discussed in (Peay, 1980; Opsahl et al, 2010).

It should be observed that these extensions are not yet widely adopted. One reason is that giving accurate estimates of edge weights requires a deep understanding of social relationships; this kind of information can be extremely difficult to infer and is often domain-specific, i.e., it depends on the type of social network. On the other hand, simple structural properties (who is connected to whom) are much easier to identify.

The third assumption is that the graph is fully connected, i.e., each pair of nodes is connected by at least one path. In fact, social networks tend to have a large connected component, which includes most of the nodes that are typically those of interest. Therefore, analysis of disconnected networks is usually carried out by first splitting the graph into its connected components, and then analyzing each component separately.

We now introduce the definition of single-node measures, including for each one a brief description and, where appropriate, some usage hints or limitations. We then illustrate the corresponding whole-network measures, which are typically derived by aggregating the single-node values. It is important to observe that some network measures have been defined by different authors in different ways. e.g., with and without normalization factors. Therefore, different network analysis tools may provide different results when computing the same measure, since they may be using different definitions.

Single-Node and Path-based Metrics

In this section we describe the most commonly used path-based and single-node measures in SNA.

Shortest Paths and Geodesic Distance

Many important graph metrics are based on the concepts of *shortest path* and *geodesic distance*. A shortest path between two nodes u and v on an unweighted graph is a

Table 1. Geodesic distance $d(u, v)$ between all pairs of nodes in the working example.

	n_0	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8
n_0	0	1	2	1	1	1	3	3	4
n_1	1	0	1	2	2	2	2	2	3
n_2	2	1	0	3	3	3	1	1	2
n_3	1	2	3	0	2	2	4	4	5
n_4	1	2	3	2	0	1	4	4	5
n_5	1	2	3	2	1	0	4	4	5
n_6	3	2	1	4	4	4	0	2	3
n_7	3	2	1	4	4	4	2	0	1
n_8	4	3	2	5	5	5	3	1	0

path from u to v with minimum number of edges; the geodesic distance $d(u, v)$ is the number of edges of one of the shortest paths from u to v . For undirected graphs, $d(u, v) = d(v, u)$ since all paths can be reversed. By definition, the distance of a node from itself is zero ($d(u, u) = 0$); if u and v are adjacent, meaning that there exists an edge between them, then $d(u, v) = 1$. If there is no path connecting u and v , we set $d(u, v) = +\infty$. Since we focus on undirected, connected graphs, we assume that no pair of nodes has infinite distance, i.e., all pairs of nodes are connected. We also assume that no edges connecting a node to itself (also called loops) exist. Table 1 shows the geodesic distance between all pairs of nodes in the graph of Figure 2.

Eccentricity

The *eccentricity* $E(u)$ of a node u is defined as the maximum distance between u and all other nodes (Harary and Norman, 1953; Harary, 1969). Formally:

$$E(u) = \max_{v \in V} d(u, v) \quad (1)$$

The eccentricity is a measure of how efficiently a node can disseminate information, with lower values denoting better dissemination efficiency. In our example, if we

Table 2. Single-node measures for the working example (values approximated to the second decimal)

Node	Eccentricity	Closeness	Betweenness	Clustering Coefficient
n_0	4	0.50	17.0	$\frac{1}{6}$
n_1	3	0.53	16.0	0.0
n_2	3	0.50	17.0	0.0
n_3	5	0.35	0.0	0.0
n_4	5	0.36	0.0	1.0
n_5	5	0.36	0.0	1.0
n_6	4	0.35	0.0	0.0
n_7	4	0.38	7.0	0.0
n_8	5	0.29	0.0	0.0

consider the geodesic distances shown in Table 1, we observe that node n_8 has eccentricity 5 (maximum value on the last row or column of the table), which means that the furthest nodes from n_8 are located 5 hops away. We can actually see that nodes n_3 , n_4 , and n_5 have geodesic distance 5 from n_8 ; this means that, for example, information produced by n_8 can pass through n_7 , n_2 , n_1 and n_0 to reach n_4 in the smallest number of steps. On the other hand, nodes n_1 and n_2 have both eccentricity 3, which means that information originating from them must traverse at least 3 edges before reaching some specific nodes in the graph. However, notice that information does not necessarily pass through the shortest paths of the network. For example, a message from n_0 may reach n_4 indirectly through n_5 even if there is an edge between n_0 and n_4 . The second column of Table 2 shows the eccentricity of all nodes in our working example.

Closeness

While eccentricity represents the maximum distance between nodes, closeness measures average distances. The *closeness* $C(u)$ of a node u has been defined in different ways, all providing the same information. The main idea is to use the inverse average distance

between u and all other nodes as a measure of how quickly we expect information produced by u to reach the rest of the network (Sabidussi, 1966). Formally:

$$C(u) = \frac{n-1}{\sum_{v \in V, v \neq u} d(u, v)} \quad (2)$$

where n is the number of nodes of the graph G . n is sometimes used as a normalization factor instead of $n-1$. Moreover, $1/C(u)$ is used in some SNA tools as a measure of closeness, although it would be more appropriate to consider it as a measure of *farness* since it increases when the average distance increases. However, all these alternative definitions return the same ranking of nodes (in inverse order, if the inverse of Eq. (2) is used), so they can be typically used interchangeably.

Note that the definition of closeness relies on the assumption that the graph is connected, so that there are $n-1$ other nodes connected with u . If we consider Figure 2, the closeness of node n_0 can be computed by averaging the distances of all other nodes from n_0 , as reported in Table 1, and inverting the result. Therefore, we get $C(n_0) = 8/(1+2+1+1+1+3+3+4) = 0.5$. The third column of Table 2 shows the closeness of each node of our sample graph.

When closeness is used to compare different nodes or networks, it is important to consider that this measure usually spans a small range of values. In real social networks the distance between nodes tends to be small, while the number of nodes can be very large. In addition, the definition of closeness according to (2) fails to take into consideration the fact that distant edges are less likely to play important roles in processes we may be interested in, such as information spreading, since in (2) all edges in a shortest path contribute equally. For example, n_0 may rely more on its neighbors n_1 , n_3 , n_4 and n_5 to forward its messages than n_7 , that is not directly connected to it. As a consequence, alternative definitions of closeness may be considered, assigning

different weights to specific edges depending on their distance from the node under examination.

Betweenness

Betweenness measures how frequently a node lies on shortest paths between other nodes (Freeman, 1977; Anthonisse, 1971). Let $\sigma_{vw}(u)$ be the number of geodesic paths (shortest paths) from v to w passing through u , and σ_{vw} the total number of geodesic paths between nodes v and w . Then, the ratio $\sigma_{vw}(u)/\sigma_{vw}$ can be interpreted as the probability that node u lies on a randomly selected geodesic path from v to w . The *betweenness* $B(u)$ of node u is defined as

$$B(u) = \sum_{\substack{v,w \in V \\ v \neq w \neq u}} \frac{\sigma_{vw}(u)}{\sigma_{vw}} \quad (3)$$

Betweenness gives a measure of the load placed on a given node. Intuitively, if a node u has a large value of betweenness then it tends to appear on many shortest paths. Since shortest paths are the most efficient way to route information, a node with high betweenness is more likely to play an important role in the dissemination of information (White and Borgatti, 1994).

Considering our example, peripheral nodes like n_8 do not belong to any geodesic path, therefore their betweenness is 0. On the other hand, let us consider node n_2 . If one of the nodes $\{n_8, n_7\}$ wants to send a message to any of $\{n_0, n_1, n_3, n_4, n_5, n_6\}$ through a geodesic path, then the message must pass through n_2 . Looking at Figure 2 it clearly appears how n_2 plays an important role in allowing information to pass from one side of the network to the other side. The fourth column of Table 2 shows the betweenness of all nodes in the graph.

It is important to observe that the definition of betweenness centrality assumes that information always flows through geodesic paths. While this may not always be the

case in real social graphs, we may consider shortest paths as the most likely information channels and thus use this definition of betweenness as an estimate of the real number of messages passing through a node. However, more complex versions of betweenness have been proposed, taking non-shortest paths into consideration (Newman, 2005).

Clustering Coefficient

The clustering coefficient (Watts and Strogatz, 1998) measures the tendency of the neighbors of a node to be connected to each other forming a fully connected sub-graph (clique). The relevance of this measure comes from the fact that real social networks present a higher clustering coefficient than corresponding random networks, indicating a tendency to create triangles. Therefore, this measure discriminates between random and social networks and highlights nodes whose neighbors are well connected to each other.

Given an undirected graph $G = (V, E)$, for each node $u \in V$ we define the *neighborhood* $N(u)$ as the set of nodes directly connected with u :

$$N(u) = \{v \in V \mid (u, v) \in E\}$$

Let $EN(u)$ be the set of edges of G whose endpoints are both in the neighborhood $N(u)$:

$$EN(u) = \{(v, w) \in E \mid v \in N(u), w \in N(u)\}$$

Let $n(u) = |N(u)|$ be the number of neighbors of node u . We observe that the maximum number of edges that an undirected graph with $n(u)$ nodes can have is $n(u) \times (n(u) - 1) / 2$; this number corresponds to the number of edges of the complete graph with $n(u)$ nodes (in a complete graph there is an edge connecting each pair of nodes).

The *clustering coefficient* $CC(u)$ of u is defined as the ratio between the number of edges in $EN(u)$ and the number of edges that the neighborhood could have if all its

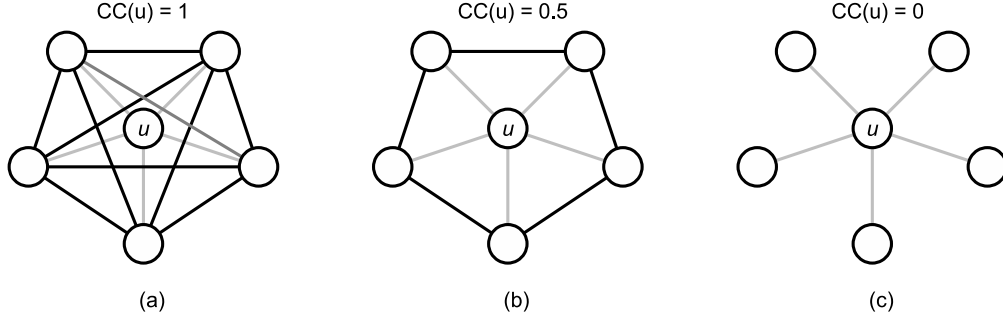


Fig. 3. Clustering coefficient for node u with different neighborhood; edges incident to u are shown in gray, since they do not contribute to the actual computation of the centrality value.

Table 3. Whole-network measures for the graph in Figure 2

Measure	Value
Diameter	5
Closeness	0.32
Betweenness	0.71
Clustering Coefficient	0.24
Density	0.25

nodes were fully connected. Formally:

$$CC(u) = \frac{2 \times |EN(u)|}{n(u) \times (n(u) - 1)} \quad (4)$$

$CC(u)$ can take values in the range $[0, 1]$. If $CC(u) = 1$, then node u together with its neighbors form a clique, as in Figure 3 (a). If $CC(u) = 0$, then the neighbors of u are not connected to each other, as in Figure 3 (c).

Whole-Network Measures

In this section we describe how the measures introduced in the previous section can be extended to whole graphs. Table 3 summarizes these measures for the graph in Figure 2.

Diameter

The graph *diameter* is a global measure of eccentricity. The diameter $D(G)$ of a graph $G = (V, E)$ is the maximum eccentricity of its nodes:

$$D(G) = \max_{u \in V} E(u) \quad (5)$$

which, applying (1), can be rewritten as:

$$D(G) = \max_{u, v \in V} d(u, v) \quad (6)$$

Therefore, the diameter of G is the maximum distance between all pair of nodes. For the graph of Figure 2, we observe from the data shown in the second column of Table 2 that the maximum eccentricity is 5; therefore, the graph diameter is 5.

The diameter of a graph is an indication of the efficiency of information transmission: information produced by a node may need to traverse $D(G)$ edges to reach other specific nodes in the network. Note that information may also traverse more than $D(G)$ edges if it does not follow shortest paths. In addition, for graphs with a dense core and a few distant nodes, the diameter is determined by those nodes and may assume a high value, whereas the majority of nodes are close to each other.

Closeness

According to Freeman (1978), the network closeness $C(G)$ of a graph G is defined as:

$$C(G) = \frac{\sum_{v \in V} (C_{max} - C(v))}{(n^2 - 3n + 2)/(2n - 3)} \quad (7)$$

where $C(v)$ is the closeness of node v , and $C_{max} = \max_{v \in V} C(v)$ is the maximum value of closeness over the whole graph. In our example we have $C(G) = 0.32$, as it can be obtained from the values in column *closeness* of Table 2.

The feature emphasized by closeness is similar to those emphasized by eccentricity and diameter. However, a single node far away from the others would affect

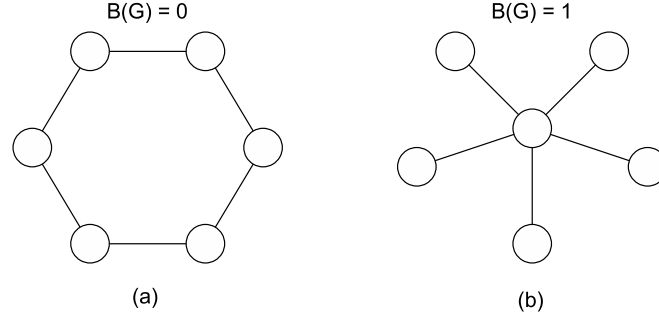


Fig. 4. Extreme values of betweenness centrality

closeness to a more limited extent than eccentricity and diameter. Also in this case a higher value of closeness corresponds to a node or network where information may tend to reach other nodes more quickly.

Other measures that are related to network closeness are the *average geodesic distance*, i.e., the mean value of the distance between all pairs of nodes, and the so-called *global efficiency* (Latora and Marchiori, 2001) $E(G)$ defined as:

$$E(G) = \frac{1}{n(n-1)} \sum_{\substack{u,v \in V \\ u \neq v}} \frac{1}{d(u,v)} \quad (8)$$

Betweenness

Betweenness has been extended by Freeman (1977) to whole graphs to represent the dominance of the “most central” node. Formally, given a graph G , let $B_{max} = \max_{v \in V} B(v)$ be the maximum value of the betweenness centrality of any node in G . The graph centrality $B(G)$ can then be defined as:

$$B(G) = \frac{\sum_{v \in V} (B_{max} - B(v))}{(n-1)B_{max}} \quad (9)$$

The value of $B(G)$ is in the range $[0, 1]$. $B(G) = 0$ if all nodes in G have the same centrality; this happens, e.g., for a graph where nodes are connected as in a ring, as shown in Figure 4 (a). The value $B(G) = 1$ can be obtained on a graph with a single central node which is connected to all other nodes in a star topology, as in Figure 4 (b).

Clustering coefficient

The *clustering coefficient* $CC(G)$ of a graph G can be computed as the average of the clustering coefficient of all nodes:

$$CC(G) = \frac{\sum_{u \in V} CC(u)}{n} \quad (10)$$

A different definition was proposed by Luce and Perry (1949). Given an undirected graph G we define a *triplet* to be a path over a set of three nodes, e.g., $\langle n_3, n_0, n_5 \rangle$. We focus on nodes connected by either two (open triplet) or three (closed triplet) edges. As an example, $\langle n_3, n_0, n_5 \rangle$ is an open triplet, while $\langle n_3, n_0, n_5, n_3 \rangle$ is a closed triplet. Therefore, the clustering coefficient $CC(G)$ can also be defined as the percentage of triplets that are closed:

$$CC(G) = \frac{\text{number of closed triplets}}{\text{total number of triplets}} \quad (11)$$

This measure has also been extended to weighted networks by Opsahl and Panzarasa (2009).

As said, when computed on a social network the clustering coefficient measures the tendency of individuals to form triangles. In fact, it is often described as a measure of the so-called *transitivity* of a graph (the term *transitivity* is sometimes used as a synonym of whole-network clustering coefficient). More in general, the clustering coefficient of a network indicates a tendency to form dense sub-graphs, also called *communities* or *clusters* depending on their interpretation.

Other worth-mentioning related measures are the *rich-club coefficient*, indicating the tendency of nodes of high degree to be well connected to each other, and the *degree correlation*, i.e., the probability that an arbitrary edge connects two nodes of specific degrees. The more general concept of *assortativity* (also known as *assortative mixing*) is used to indicate the tendency of nodes to connect to similar nodes, and one well-known assortativity measure is *modularity*, used by several methods of community

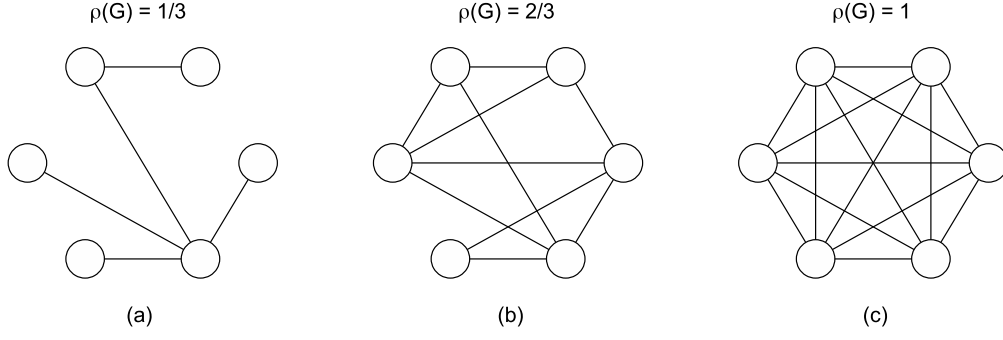


Fig. 5. Graph density

detection (Fortunato, 2010). Additional details on these measures can be found for example in (Menczer et al, 2020; Newman, 2018).

Density

The *density* $\rho(G)$ of a graph G is the fraction of edges in G with respect to the maximum number of possible edges. For an undirected graph with n nodes there can be at most $n(n-1)/2$ edges, therefore:

$$\rho(G) = \frac{2m}{n(n-1)} \quad (12)$$

where $m = |E|$ is the number of edges of G .

Density is always in the range $[0, 1]$. If $\rho(G) = 0$, then G has no edges and all nodes are isolated; if $\rho(G) = 1$ then G is a complete graph, where every pair of nodes is connected by an edge. Note that the minimum density for a connected graph with n nodes is $2/n$, since any connected graph must have at least $n-1$ edges. The converse is not necessarily true: a graph with $n-1$ edges may be disconnected.

Figure 5 shows the density of a graph with $n = 6$ nodes, with increasing number of edges. The graph in Figure 5 (a) has $m = 5$ edges, resulting in a density $\rho(G) = (2 \times 5)/(6 \times 5) = 1/3$. The graph in Figure 5 (b) has $m = 10$ edges, resulting in a density $\rho(G) = (2 \times 10)/(6 \times 5) = 2/3$. Finally, the graph in Figure 5 (c) is fully connected and therefore has density $\rho(G) = 1$.

Table 4. Notation Summary

V	Set of graph nodes
E	Set of graph edges, $E \subseteq V \times V$
n	Number of nodes
m	Number of edges
$d(u, v)$	Geodesic distance from node u to node v
σ_{vw}	Number of shortest paths from v to w
$\sigma_{vw}(u)$	Number of shortest paths from v to w passing through u
$N(u)$	Neighbors of node u
$n(u)$	Number of neighbors of node u , $n(u) = N(u) $
$EN(u)$	Set of edges in the neighborhood of u

The density of social graphs is typically low, meaning that on average every individual is connected to a small number of other individuals. This can be observed on any online social network site, e.g., Facebook or X/Twitter, where most users are connected to tens or hundreds of other users out of about a billion total users. This feature can be explained by the existence of limits in cognitive activity preventing a person to manage more than a given number of stable relationships (Goncalves et al, 2011) and is fundamental for the design of efficient algorithms. For our graph in Figure 2 we have $n = 9$ nodes and $m = 9$ edges, resulting in a density of $\rho(G) = (2 \times 9)/(9 \times 8) = 1/4$.

Summary

We summarize the notation used in this essay in Table 4. Table 5 lists all single-node and whole-network measures introduced so far.

Table 5. Single-Node and Whole-Graph metrics

Single-Node		Whole-Graph	
Eccentricity	$E(u) = \max_{v \in V} d(u, v)$	Diameter	$D(G) = \max_{u \in V} e(u)$
Closeness	$C(u) = \frac{n-1}{\sum_{v \in V, v \neq u} d(u, v)}$	Closeness	$C(G) = \frac{\sum_{v \in V} (C_{max} - C(v))}{(n^2 - 3n + 2)/(2n - 3)}$
Betweenness	$B(u) = \sum_{\substack{v, w \in V \\ v \neq w \neq u}} \frac{\sigma_{vw}(u)}{\sigma_{vw}}$	Betweenness	$B(G) = \frac{\sum_{v \in V} (B_{max} - B(v))}{(n-1)B_{max}}$
Clust. Coefficient	$CC(u) = \frac{2 \times EN(u) }{n(u) \times (n(u) - 1)}$	Clustering	$CC(G) = \frac{\sum_{u \in V} CC(u)}{n}$
		Density	$\rho(G) = \frac{2m}{n(n-1)}$

Computational Aspects

In this section we describe the algorithms that can be used to compute some of the metrics described in the previous sections. We assume that the reader is familiar with the basic concepts of algorithm design and analysis, as can be found in introductory textbooks such as the one by Cormen et al (2022). However, in order to make this essay self-contained, we summarize the main points below.

An algorithm is a finite sequence of steps describing an effective method for calculating a function. A fundamental attribute of an algorithm is its *efficiency*, representing the amount of resources (e.g., CPU time or storage space) it needs to compute the result. The amount of resources depends on the input size: it is reasonable to expect that for larger inputs the algorithm will require more CPU time and/or storage space to compute the result. The input size of most graph algorithms is the size of the input graph, which in turn is proportional to the number of nodes n and edges m .

The goal of algorithm analysis is to define a function mapping the input size to the number of elementary steps (*time complexity*) or storage locations (*space complexity*) required by the algorithm to compute the result. Since it is in general not possible

to give a precise definition of the complexity function, it is common to estimate its asymptotic behavior, that is the growth rate of the complexity function for arbitrarily large inputs.

The Big O notation is used to concisely express the asymptotic growth rate of the cost function. Assume that an algorithm requires $T(n)$ steps (storage locations) to compute the results for an input of size n . We say that the asymptotic cost of the algorithm is $O(f(n))$ if there exist constants $c > 0$ and $n_0 > 0$ such that, for all $n \geq n_0$, $T(n) \leq cf(n)$. For example, we say that an algorithm requires time $O(n^2)$ if its cost function grows not faster than the square of the input size, for sufficiently large inputs.

Graph Representation

We first introduce the data structures which can be used to represent graphs. A simple way to represent a (directed or undirected) unweighted graph $G = (V, E)$ with n nodes is using an $n \times n$ *adjacency matrix* M_{ij} , where $M_{ij} = 1$ if and only if $(i, j) \in E$. For the graph of Figure 2 the adjacency matrix is:

$$\begin{array}{c}
 \begin{matrix} & n_0 & n_1 & n_2 & n_3 & n_4 & n_5 & n_6 & n_7 & n_8 \end{matrix} \\
 \begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \\ n_6 \\ n_7 \\ n_8 \end{matrix} \begin{pmatrix}
 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
 \end{pmatrix}
 \end{array}$$

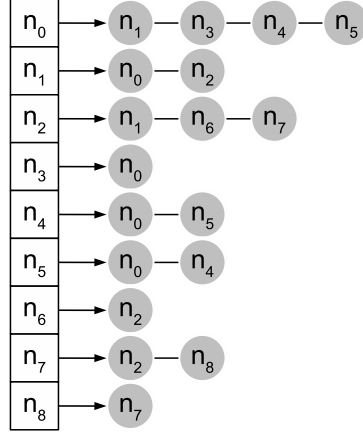


Fig. 6. Adjacency list representation of the graph on Figure 2

If G is undirected, as in the case of Figure 2, the adjacency matrix is symmetric since edges (u, v) and (v, u) are the same. Adjacency matrices support some graph operations efficiently: for example, it is possible to test for the existence of an edge (u, v) , add a new edge, or delete an edge in constant time by accessing the appropriate element of the matrix. Unfortunately, the storage space required to encode a graph with n nodes is $O(n^2)$ assuming that all the elements in the matrix are stored, since the matrix has n^2 elements. Therefore, the adjacency matrix is mostly used with small graphs or with dense graphs where $\rho(G) \approx 1$. Sparse matrices can be represented in compressed format to save space, but this would make access operations more costly.

A more space-efficient representation is based on *adjacency lists*. Here, the graph is represented as an array of n lists, where list u contains the neighbors of node u . Figure 6 shows the adjacency list representation of the graph in Figure 2. Since the graph has $n = 9$ nodes, we have 9 lists, each one associated to a single node. The list for node n_0 contains four elements $\{n_1, n_3, n_4, n_5\}$, which are precisely the four neighbors of n_0 . Note that, for undirected graphs, each edge appears twice in the lists. For example, edge (n_0, n_5) appears as element n_5 in the list for n_0 , and as element n_0 in the list for node n_5 . Therefore, the total space requirement of an adjacency list is $n + 2m = O(n + m)$. While the space requirement is lower than the adjacency

Table 6. Computational costs of adjacency matrix and adjacency list graph representation

	Adjacency Matrix	Adjacency Lists
Space	$O(n^2)$	$O(n + m)$
Checking Adjacency	$O(1)$	$O(\max_v n(v))$
List adjacent nodes	$O(n)$	$O(\max_v n(v))$
Adding an edge	$O(1)$	$O(1)$
Deleting an edge	$O(1)$	$O(\max_v n(v))$

matrix, some operations are less efficient on adjacency lists. For example, to test for the existence of an edge (u, v) , it is necessary to scan the adjacency list for u (or v) until either the other node is found, or the end of the list is reached. The cost in this case is $O(\max_v n(v))$, that is, in the worst case proportional to the maximum number of neighbors a node can have.

Table 6 shows the space complexity and time complexity of operations on adjacency matrix and adjacency list graph representations.

Graph Algorithms

We now give an overview of some classic algorithms used to compute the measures considered in this essay. The algorithms considered here are not necessarily the most efficient ones available, but are probably easier to understand.

Eccentricity, Closeness and Diameter. To compute the eccentricity and closeness of a node u we need to compute the geodesic distances from u to all other nodes. This is the well known Single Source Shortest Path (SSSP) problem on graphs (Festa, 2006). For unweighted graphs, this problem reduces to performing a Breadth First visit of the graph starting from u , that requires time $O(n + m)$ using adjacency lists. Dijkstra's algorithm can solve the SSSP problem for directed graphs with nonnegative edge weights in $O((n + m) \log n)$ time using a priority queue implemented with a binary heap (Cormen et al, 2022); the running time can be reduced to $O(m + n \log n)$ using

more efficient priority queue implementations. For directed graphs with arbitrary edge weights, Bellman-Ford algorithm can be used to compute all shortest paths from a single source in $O(nm)$ time (Cormen et al, 2022). Computing the diameter requires solving the All Pairs Shortest Path problem, that is, computing the geodesic distances between all pairs of nodes. For unweighted graphs this can be achieved in $O(n^2 + nm)$ time by executing n Breadth First visits, starting from each node. Floyd-Warshall algorithm (Floyd, 1962) can compute all geodesic distances for directed graphs with arbitrary edge weights in $O(n^3)$ time. Johnson’s algorithm (Johnson, 1977) has a computational complexity of $O(nm + n^2 \log n)$, which is more efficient for graphs with low density.

Betweenness. The most efficient sequential algorithm for computing the betweenness centrality of graphs is due to Brandes (2001). Brandes’ algorithm requires $O(n + m)$ space and runs in $O(nm)$ time on unweighted graphs, and $O(nm + n^2 \log n)$ time on weighted ones.

Clustering Coefficient. Computation of the node clustering coefficient $C(u)$ can be done in $O(n^2)$ time in the worst case, by counting all edges incident to the neighbors of u . The network clustering coefficient $C(G)$ from Eq. (11) can be computed by counting all (closed) triplets. A brute-force approach is to examine each combination of nodes, which requires $O(n^3)$ time. A better algorithm has been proposed by Latapy (2008), who demonstrated that it is possible to solve triplet finding, counting, and node-counting in $O(n^{2.376})$ time and $O(n^2)$ space using fast matrix multiplication on the adjacency matrix representation of G .

Software

We conclude this part by mentioning some available software packages which can be used to compute the measures described above and many others.

Gephi (Bastian et al, 2009) is an interactive network visualization tool supporting visual network analysis. **Gephi** is a cross-platform and extensible environment, with a plugin mechanism to implement additional algorithms; it provides algorithms to compute common SNA measures, including those addressed in this essay. **igraph** (Csardi and Nepusz, 2006) is a software package written in C for creating and manipulating large graphs, and can be used for statistical SNA thanks to its version for the R statistical environment (R Core Team, 2021). **NetworkX** (Hagberg et al, 2008) is a software package written in Python that includes algorithms for creating, manipulating and analyzing graphs. **NetworkX** can generate directed or undirected graphs, with or without self-loops and parallel edges; several types of static and random graph generators are included. **NetworkX** allows the computation of a fairly comprehensive list of metrics, exceeding those that have been discussed in this essay.

Finally, we mention **GraphBLAS** (Brock et al, 2023) which is not a software package but an API for representing and manipulating graphs using linear algebra transformations, e.g., vector-matrix operations. At the time of writing, there exist C/C++/Java implementations that conform to the **GraphBLAS** interface.

Key Applications

The applications of Social Network Analysis span the most diverse topics. Newman (2018) cites several examples, including the estimation of the number of people the average person knows (McCormick et al, 2010), the study of the collaboration pattern of scientists (Newman, 2001) and movie actors (Watts and Strogatz, 1998), the analysis of dating patterns among high school students (Bearman et al, 2004), and the analysis of networks of terrorists (Latora and Marchiori, 2004). We briefly discuss some of these results.

A scientific collaboration network is an undirected graph $G = (V, E)$ where nodes represent scientists, and there exists an edge (u, v) if and only if u and v wrote a paper together. Newman (2001) observed that the collaboration networks of several disciplines exhibit a *small-world* structure: two scientists picked at random are likely to be connected by a short path in G . Specifically, the average degree of separation in the analyzed dataset is about 6, meaning that any scientist can be reached from any other scientist in the collaboration graph by following a path of average length 6.

The small-world property of collaboration networks has been part of the folklore for a long time before being formally observed. Mathematicians defined the Erdős number as a tribute to Paul Erdős (1913–1996), probably the most prolific mathematician of all times. The Erdős number $E(v)$ is the distance between Erdős and researcher v in the collaboration graph. Therefore, Erdős himself has Erdős number 0; those who have written a paper with Erdős have $E(v) = 1$; their coauthors (that are not coauthors of Erdős) have $E(v) = 2$, and so on. Most mathematicians have finite Erdős number: an estimate of the median Erdős number among mathematicians is 5, the mean is 4.65 and the standard deviation is 1.21 (Erdős number project, 2024). Many non-mathematicians have finite Erdős number as well, due to inter-disciplinary research activities that resulted in joint publications at some point in time.

The idea of Erdős number has been ported to the movie industry: the *Bacon number* is defined as the distance to Kevin Bacon in the undirected graph G whose nodes are actors, and an edge (u, v) represents the fact that actors u and w appeared in the same movie. The average Bacon number for a randomly chosen actor with finite distance to Kevin Bacon is 3.02 (Bacon Oracle, 2024), revealing that the movie industry is a small-world network. There are notable scientists that have both finite Erdős number and finite Bacon number: for example, physicist and Nobel laureate Richard

Feynman has Erdős number 3 and Bacon number 3, the latter due to his appearance in the film *Anti-Clock*.

Humans are not the only specie whose collaboration patterns take the form of a small-world network. Lusseau (2003) analyzed the social interaction graph within a community of 64 bottlenose dolphins (*Tursiops truncatus*). Nodes of the interaction graph correspond to individual dolphins, and an undirected edge (u, v) denotes that u and v were observed together more often than expected by random encounters alone. The resulting network is scale-free and highly clustered.

We conclude by describing a study where graph metrics have been used to probe the structure of a terrorist organization. Latora and Marchiori (2004) consider the connections among the hijackers involved in the September 2001 attacks, with the goal of identifying the terrorists to target if one wants to disrupt the organization. As a quantitative measure of such disruption, the authors use the global efficiency of the terrorist graph defined in Eq. (8); the most important individual in the criminal organization is the one whose removal produces the largest decrease in efficiency.

Future Directions

The success of online social network sites as a mass phenomenon has dramatically increased the size of these networks. Analyzing them is computationally challenging, due to the prohibitive time and storage space requirements of traditional, sequential algorithms. In response to these challenges, new algorithms have been proposed, based on emerging computer architectures and different computing paradigms like approximate algorithms (Brandes and Pich, 2007) and streaming computation (Becchetti et al, 2008; Guha and McGregor, 2012). On-chip parallelism, such as that provided by modern multicore processors or by general-purpose Graphics Processing Units (GPUs), is fostering the interest on the development of efficient parallel graph algorithms (Lamber-

tini et al, 2014; Lumsdaine et al, 2007; Wang et al, 2016), where multiple independent execution units concurrently and cooperatively build the solution to specific graph problems. The research activity in this area has lead to development of several software packages for network analysis on multicore/manycore processors; among others, **NetworKit** (Angriman et al, 2022) and **graph-tool** (Peixoto, 2014) are Python packages that rely on algorithms implemented in C++ that can use multiple CPU cores, while **cugraph** (Fender et al, 2022) relies on GPUs.

However, it should be observed that more powerful algorithms and computing infrastructures are not always the correct solution to address the increase in network sizes. For example, while it may be interesting to compute centrality measures on the whole Facebook network, practical SNA tasks would often focus on a specific community, e.g., people subscribed to a product or company page or fans of a public figure. In this context community detection methods may become particularly relevant to filter portions of the network to analyze (Fortunato, 2010).

Another aspect that has to be considered when constructing social networks from complex data sources is the semantics of edges. Some decades ago the reconstruction of the social graph was a difficult task, usually to be performed by asking people to list their friends or observing specific environments, e.g., a workplace. The advent of on-line social networks has made it much easier to construct social network graphs. However, the graph edges that can be easily inferred from online data may not be the best way to identify the social relations of interest, and this could lead to the construction of a wrong social graph. For example, many social network sites provide information about user contacts through their Application Programming Interfaces (API), e.g., we can easily retrieve all X/Twitter followers of a specific user through the X/Twitter API. However, all the path-based measures presented in this essay are based on the assumption that the network represents communication paths, while a

large percentage of messages is in fact exchanged between users that are not directly connected, and therefore they are exchanged on a different network that we may call *communication network* (Rossi and Magnani, 2012).

Things may get even more complicated when we consider multiple social networks. Individuals often use different services to communicate with different audiences, e.g., Facebook, X/Twitter or LinkedIn. Of course, the social graphs built using Facebook, X/Twitter or LinkedIn contacts may differ strongly, even if these graphs have the same set of individuals. The computation of centrality measures on each graph in isolation may provide insights on the usage of that social network by a specific user, but may fail to draw a realistic picture of the information flow and of the role each user plays in cross-disseminating information over all networks. Multilayer social networks can be used to model and measure systems of multiple networks (Magnani and Rossi, 2011; Dickison et al, 2016).

Networks can also be enriched in other ways to provide more accurate models of reality. For example, links can be annotated with probabilities, to represent our uncertainty about their existence coming from imperfect data collection processes. Path-based measures can also be computed for such probabilistic networks: instead of a single distance between two nodes, one can compute a probability distribution over the set of possible distances. A distribution of possible distances can then be summarised using common statistical summaries such as the median distance (Potamias et al, 2010). We may decide to retain only the shortest paths whose probability exceeds a given threshold (Yuan et al, 2010), or compute the most reliable path, that is, the path with the highest probability (Sevon et al, 2006). Measures based on probabilistic paths, such as betweenness and closeness, can then also be defined (Kaveh et al, 2020). Other rich network models include temporal networks (Holme and Saramäki, 2023) and temporal text networks (Vega and Magnani, 2018, 2023), including information about the time

when links exist and (for the latter) also the text that is exchanged between nodes. Extended path-based measures can be defined for these and other network models going beyond nodes and simple links.

Cross-references

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- Communities Discovery and Analysis in Online and Offline Social Networks (00006)
- Gephi (00299)
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